

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****CERTIFICATE OF INSTALLATION****Note:** This table completed by **HERSECC** Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. Roof/Ceiling Insulation

01	02	03	04	05	06	07	08	09	10
I.D.	Manufacturer & Brand	Assembly/ Framing Material	Assembly Thickness (inches)	Framing Size & Spacing	Insulation Type	Ceiling Insulation R-value	Insulation Depth (inches)	Continuous Insulation Above the Roof Deck R-value	Insulation Below the Roof Deck R-value

B. Wall Insulation

01	02	03	04	05	06	07	08	09	10
I.D.	Manufacturer & Brand	Assembly/ Framing Material	Assembly Thickness (inches)	Framing Size & Spacing	Insulation Type	Core/Cavity Insulation R-value	Insulation Depth (inches)	Continuous Exterior Insulation R-value	Continuous Interior Insulation R-value

C. Masonry/Mass Insulation

01	02	03	04	05	06	07	08	09
I.D.	Manufacturer & Brand	Walls Above Grade	Mass Thickness (inches)	Exterior Furring Strip Type/ Depth (inches)	Interior Furring Strip Type/Depth (inches)	Insulation Type	Exterior Insulation R-value	Interior Insulation R-value

D. Raised Floor Insulation

01	02	03	04	05	06	07	08	09
I.D.	Manufacturer & Brand	Framing Material	Framing Size & Spacing	Insulation Type	Cavity Insulation R-value	Insulation Depth (inches)	Exterior Floor Insulation R-value	Concrete Fill

E. Slab Floor/Perimeter Insulation (See Section F. for Insulation Requirements for Heated Slabs)

01	02	03	04	05	06	07	08
I.D.	Manufacturer & Brand	Floor Type	Insulation Type	Insulation Depth (inches)	Insulation R-Value	Vertical Insulation Length (inches)	Horizontal Insulation Length (feet)

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****F. Heated Slab Insulation**

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	All heated slabs shall be insulated as required by Section 110.8(g). Footings must meet required insulation levels.
02	Insulation shall be installed from the top of the slab, down 16 inches or to the frost line, whichever is greater. Climate zones 1-15 requires R-5, climate zone 16 requires R-10.
03	Alternatively, vertical insulation from top of slab at inside edge of outside wall down to the top of the horizontal insulation. Horizontal insulation from the outside edge of the vertical insulation extending 4 feet toward the center of the slab in a direction normal to the outside of the building in plain view. Climate zones 1-15 require R-5, and climate zone 16 requires R-10 vertical and R-7 horizontal.

G. Minimum Mandatory Measures

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Insulation - 110.8(a): All installed insulation is certified and listed with the Department of Consumer Affairs, Bureau of Household Goods and Services.
02	Insulation - 110.8(b)2: Urea formaldehyde foam insulation is protected by 4 mil polyethylene vapor retarder.
03	Insulation - 110.8(c): Flame spread and smoke density requirements of the California Building Code (CBC) are met.
04	Raised Floor - 150.0(d): All raised wood-frame floors have a minimum R-19 insulation or equivalent U-factor.
05	Slab Floor/Perimeter - 150.0(f): Water absorption rate for the insulation material alone without facings is no greater than 0.3%; water vapor permeance rate is no greater than 2.0 perm/inch, and perimeter insulation is protected from physical damage and UV light deterioration.
06	Above Grade Exterior Wall - 150.0(c)1 & 150.0(c)6: All 2x4 wood-frame walls have a minimum R-13 insulation or equivalent U-factor not exceeding U-0.095402.
07	Above Grade Exterior Wall - 150.0(c)2 & 150.0(c)6: All 2x6 wood-frame walls have a minimum R-20 insulation or equivalent U-factor not exceeding U-0.069074.
08	Roof Deck - 150.0(a)1: All newly constructed attics <u>that are above conditioned space systems</u> have a maximum area-weighted average U-factor not exceeding U-0.184.
09	Ceiling/Rafter Roof - 150.0(a)2: All wood-framed ceilings have a minimum R-22 insulation or equivalent U-factor.
10	Vapor Retarder – 150.0(g)1: In Climate Zones 1 through 16, the earth floor of unvented crawl space shall be covered with a Class I or Class II vapor retarder. This requirement shall also apply to controlled ventilation crawl space for buildings complying with the Exception to Section 150.0(d).
11	Vapor Retarder – 150.0(g)2: In Climate Zones 14 and 16, a Class I or Class II vapor retarder shall be installed on the conditioned space side of all insulation in all exterior walls, vented attics and unvented attics with air-permeable insulation.
12	Heated Slabs - 110.8(g): All heated slabs shall be insulated as required. - Insulation shall be installed from the top of the slab, down 16 inches or to the frost line, whichever is greater. Climate zones 1-15 require R-5, and climate zone 16 requires R-10. - Alternatively, vertical insulation from top of slab at inside edge of outside wall down to the top of the horizontal insulation. Horizontal insulation from the outside edge of the vertical insulation extending 4 feet toward the center of the slab in a direction normal to the outside of the building in plain view. Climate zones 1-15 require R-5, and climate zone 16 requires R-10 vertical and R-7 horizontal.

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****H. Installed Insulation**

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Installed insulation R-values are the same or greater than listed on the CF1R.
02	No gaps or voids between the insulation and framing.
03	No gaps between the sides or ends of batt insulation.
04	Loose-fill insulation must be installed to the minimum installed weight per square foot (density) of the manufacturer's cut sheet for the proposed R-value.
05	Batt insulation is not compressed (no stuffing of the insulation into the cavity) and is installed to its full thickness.
06	Insulation is cut around obstructions such as electrical boxes.
07	Batt insulation is delaminated around all plumbing and electrical lines in ceilings, walls, and floors.
08	Band joists are insulated to the same R-value as the wall.
09	In all narrow cavities the insulation shall be cut to fit or filled with expanding foam.
10	Insulation was installed per manufacturer instructions.

I. Wall Insulation

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	When allowed by the manufacturer, low expanding foam shall be used to fill gaps and voids around windows and doors. If not, the cavity must be airtight and filled completely with insulation. Batt insulation must be cut to width. No stuffing allowed.
02	Install wall insulation before installing tubs, showers, and fireplaces.
03	Electric panels on walls separating conditioned and unconditioned space are sealed and insulated behind the panel with rigid insulation or expanding foam.
04	All walls of interior closets vented to the outside for HVAC or water heating equipment have the same R-value and air barrier as the exterior walls and ceiling. Doors are insulated and weather stripped.
05	Ducting is not allowed in exterior walls unless it is insulated to R-6 or greater, and the insulation and ducting are not crushed. .
06	Corner channels, wall intersections, and double sided shear walls are insulated to the required R-value before enclosing the wall.
07	Insulation that does not fill the cavity is placed against the exterior air barrier.
08	Band joists are insulated to the same R-value as the walls.

J. Ceiling/Roof Insulation

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Insulation extends to the outside edge of the exterior top plates and is flush against any ventilation dams/baffles.
02	Insulation is in direct contact with ceiling, so there are no gaps between the ceiling and the insulation.
03	For chimneys and flues, the insulation is in contact with the sheet metal collar.
04	Can lights are covered with insulation to the same depth as required by the CF1R for ceiling insulation. If not, an area weighted calculation is required to be turned in with this compliance document (CF1R-ENV-02-E).
05	Walkways and mechanical platforms are insulated to the same R-value as required for the ceiling. If not, an area weighted calculation is required to be turned in with this compliance documents (CF1R-ENV-02-E).
06	Insulate soffits by adding an air barrier and covering with insulation, or insulate the entire soffit including floor and walls.
07	Knee walls and skylight shafts are insulated to the wall R-value and in full contact with the interior air barrier. If framing on these surfaces is laid flat batt insulation is cut to fit around the framing. Batt insulation is not allowed to be draped over the framing.
08	Attic access doors are insulated to the same R-value as the ceiling. The insulation is permanently attached using adhesive or mechanical fasteners.
09	Attic access must be surrounded with a dam at least the same depth as the insulation to prevent loss of ceiling insulation.
10	Batt insulation is cut to fit around cross bracings and truss webs in the attic.

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****K. Raised Floor Insulation**

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Insulation is in full contact with subfloor.
02	Insulation hangers are spaced at 18 inches or less; insulation hangers must not compress insulation.
03	If netting or mesh is used, the cavity under the floor is filled and in contact with the subfloor.
04	If the basement is conditioned, the walls adjacent to the crawlspace must meet minimum wall R-value requirements. This includes framed stem walls, and vertical concrete retaining walls.
05	If access to the crawl space is from the conditioned area, the raised floor must have an airtight insulated access hatch.

L. Floor Above Garage Insulation

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Insulation must be in full contact with the subfloor if the air barrier is at the band joist at the garage/house wall.
02	Insulation hangers spaced at 18 inches or less; insulation hangers must not compress insulation.
03	If netting or mesh is used, the cavity under the floor is filled and in contact with the subfloor.
04	If the air barrier is at the perimeter of the garage, below the conditioned subfloor, the insulation is placed on the garage ceiling. The perimeter of the subfloor is also insulated.

M. Cantilevered Floor Insulation

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	Insulation is in full contact with the cantilevered subfloor. Insulation hangers are spaced at 18 inches or less; insulation hangers do not compress insulation.
02	If netting or mesh is used, the cavity under the floor is filled and in contact with the subfloor.
03	Sealed blocking is installed between joists where a wall rim joist would be located in the absence of a cantilever. Insulation is placed on both sides of this block.

N. Attached Porch Roof Insulation

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

01	The exterior insulated wall at the intersection with the porch roof is fully insulated above, below, and behind the roof line.
02	Where truss framing is used, airtight blocking is installed at the top and bottom of each wall/roof section and insulated.

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****DOCUMENTATION AUTHOR'S DECLARATION STATEMENT**

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/ HERSECC Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Installation is true and correct.
2. I am either: a) a responsible person eligible under Division 3 of the Business and Professions Code in the applicable classification to accept responsibility for the system design, construction, or installation of features, materials, components, or manufactured devices for the scope of work identified on this Certificate of Installation, and attest to the declarations in this statement, or b) I am an authorized representative of the responsible person and attest to the declarations in this statement on the responsible person's behalf.
3. The constructed or installed features, materials, components or manufactured devices (the installation) identified on this Certificate of Installation conforms to all applicable codes and regulations and the installation conforms to the requirements given on the Certificate of Compliance, plans, and specifications approved by the enforcement agency.
4. I understand that a registered copy of this Certificate of Installation shall be posted or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
5. I understand that a registered copy of this Certificate of Installation is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone:	Date Signed:

CF2R-ENV-03-E User Instructions

A. Roof/Ceiling Insulation

1. I.D.: A label from the plans (e.g., A1.4 or Roof) documenting the location of the installed insulation.
2. Manufacturer and Brand: Indicate the manufacturer and brand of the product being installed.
3. Assembly/Framing Material: Wood, Metal, SIP OSB, SIP I-Joist, SIP Single 2x, SIP Double 2x, see JA4 for guidance.
4. Thickness: Thickness in inches.
5. Framing Size & Spacing: Indicate the framing size and spacing (e.g., 2x4 @ 16 in O.C.).
6. Insulation Type: Using the drop down menu, select the type of insulation being installed (e.g., cellulose, fiberglass, SPF, etc.).
7. Ceiling Insulation R-value: Indicate the ceiling insulation R-value.
8. Insulation Depth: Indicate, in inches, the amount of insulation installed.
9. Continuous Insulation Above the Roof Deck R-value: Indicate the R-value of continuous insulation, having no framing penetration, installed above the roof deck.
10. Insulation Below the Roof Deck R-value: Indicate the R-value of insulation installed below the roof deck.

B. Wall Insulation

1. I.D.: A label from the plans, (e.g., A1.4 or Wall1) documenting the location of the installed insulation.
2. Manufacturer and Brand: Indicate the manufacturer and brand of the product being installed.
3. Assembly/Framing Material: Wood, Metal, SIP OSB, SIP I-Joist, SIP Single 2x, SIP Double 2x, see JA4 for guidance.
4. Thickness: Thickness in inches.
5. Framing Size & Spacing: Indicate the framing size and spacing (e.g., 2x4 @ 16 in O.C.).
6. Insulation Type: Using the drop down menu, select the type of insulation being installed (e.g., cellulose, fiberglass, SPF, etc.).
7. Core/Cavity Insulation R-value: Indicate the cavity insulation R-value.
8. Insulation Depth: Indicate, in inches, the amount of insulation installed.
9. Continuous Exterior Insulation R-value: Indicate the R-value of continuous insulation, having no framing penetration, installed on the exterior.
10. Continuous Interior Insulation R-value: Indicate the R-value of continuous insulation, having no framing penetration, installed on the interior

C. Masonry/Mass Insulation

1. I.D.: A label from the plans (e.g., A1.4 or Wall1) documenting the location of the installed insulation.
2. Manufacturer and Brand: Indicate the manufacturer and brand of the product being installed.
3. Walls Above Grade: Using the down menu, select “Yes” if the mass wall is above grade.
4. Mass Thickness: Indicate the thickness of the mass, in inches, the insulation is applied to.
5. Exterior Furring Strip Type/Depth: Indicate the type and thickness of furring material installed, such as wood/1.0 inch thick.
6. Interior Furring Strip Type/Depth: Indicate the type and thickness of furring material installed, such as wood/1.0 inch thick.
7. Insulation Type: Using the down menu, select the type of insulation being installed (e.g., cellulose, fiberglass, SPF, etc.).
8. Exterior Insulation R-Value: Indicate the R-value of the insulation installed on the outside of the assembly.
9. Interior Insulation R-Value: Indicate the R-value of the insulation installed on the inside of the assembly.

D. Raised Floor Insulation

1. I.D.: A label from the plans (e.g., A1.4 or Floor1) documenting the location of the installed insulation.
2. Manufacturer and Brand: Indicate the manufacturer and brand of the product being installed.
3. Framing Material: Wood or Metal.
4. Framing Size & Spacing: Indicate the framing size and spacing (e.g., 2x4 @ 16 in O.C.).
5. Insulation Type: Using the drop down menu, select the type of insulation being installed (e.g., cellulose, fiberglass, SPF, etc.).
6. Cavity Insulation R-value: Indicate the cavity insulation R-value.
7. Insulation Depth: Indicate, in inches, the amount of insulation installed.
8. Exterior Floor R-Value: Indicate the R-value of continuous insulation, having no framing penetration, installed on the outside of the floor.
9. Concrete Fill: Indicate if the floor has a concrete fill; yes/no.

E. Slab Floor/Perimeter Insulation

1. I.D.: A label from the plans (e.g., A1.4 or Slab Floor1) documenting the location of the installed insulation.
2. Manufacturer and Brand: Indicate the manufacturer and brand of the product being installed.
3. Floor Type: Indicate the type of floor type the insulation is being applied to; such as Heated Slab or Slab on Grade.
4. Insulation Type: Using the drop down menu, select the type of insulation being installed (e.g., cellulose, fiberglass, SPF, etc.).
5. Insulation Depth: Indicate, in inches, the depth of insulation installed. Refer to F02 for additional information.
6. Insulation R-Value: Indicate the insulation R-value being installed vertically and horizontal horizontally (if applicable).
7. Vertical Insulation Length: Indicate, in inches, the length of the insulation being installed. Refer to F03 for additional information on installing both vertical and horizontal slab insulation
8. Horizontal Insulation Length: Indicate, in feet, the length of the insulation installed from the outside edge of the vertical insulation to the center of the slab.

A. Roof/Ceiling Insulation

<<if CF1R contains roof/ceiling entry, then display this section; else display section header and standard “This Section Does Not Apply” message>>

01	02	03	04	05	06	07	08	09	10
I.D.	Manufacturer & Brand	Assembly/ Framing Material	Assembly Thickness (inches)	Framing Size & Spacing	Insulation Type	Ceiling Insulation R-value	Insulation Depth (inches)	Continuous Insulation Above the Roof Deck R-value	Insulation Below the Roof Deck R-value
<<pull from CF1R>>	<<user input, text, maximum 28 characters>>	<<pull. From CF1R; else NA>>	<<pull from CF1R; else NA>>	<<pull from CF1R; else NA>>	<<User selects from list: Cellulose, Cement EPS compound, Expanded polystyrene, extruded polystyrene, Fiberglass, Insulation other, Mineral wool, Plastic polyisocyanurate, Plastic polystyrene, Polyurethane, Spray Foam Insulation (SPF), Urea formaldehyde, None>>	<<pull from CF1R but user can override with value equal to or greater than CF1R >>	<<User Input: Number; else if A08=NA then display NA>>	<<pull from CF1R (string or DecimalNonnegative) but user can override with value equal to or greater than CF1R; else NA >>	<<pull from CF1R (string or DecimalNonnegative) but user can override default value equal to or greater than CF1R; else NA >>

B. Wall Insulation

<<if CF1R contains wall entry, then display this section; else display section header and standard “This Section Does Not Apply” message>>

01	02	03	04	05	06	07	08	09	10
I.D.	Manufacturer & Brand	Assembly/ Framing Material	Assembly Thickness (inches)	Framing Size & Spacing	Insulation Type	Core/Cavity Insulation R-value	Insulation Depth (inches)	Continuous Exterior Insulation R-value	Continuous Interior Insulation R-value
<<pull from CF1R>>	<<user input, text, maximum 28 characters>>	<<pull. From CF1R>>	<<pull from CF1R; else NA>>	<<pull from CF1R>>	<<User selects from list: Cellulose, Cement EPS compound, Expanded polystyrene, extruded polystyrene, Fiberglass, Insulation other, Mineral wool, Plastic polyisocyanurate, Plastic polystyrene, Polyurethane, Spray Foam Insulation (SPF), Urea formaldehyde>>	<<pull from CF1R but user can override with value equal to or greater than CF1R >>	<<User Input: Number; else NA >>	<<pull from CF1R (string or DecimalNonnegative) but user can override with value equal to or greater than CF1R; else NA >>	<<pull from CF1R (string or DecimalNonnegative) but user can override with value equal to or greater than CF1R; else NA>>

C. Masonry/Mass Insulation

<<if CF1R contains mass entry, then display this section; else display section header and standard this section does not apply message>>

01	02	03	04	05	06	07	08	09
I.D.	Manufacturer & Brand	Walls Above Grade	Mass Thickness (inches)	Exterior Furring Strip Type/ Depth	Interior Furring Strip Type / Depth	Insulation Type	Exterior Insulation R-value	Interior Insulation R-value
<<pull from CF1R>>	<<user input, text, maximum 28 characters>>	<<pull. From CF1R; else allow user to select: Yes or No>>	<<pull from CF1R>>	<<user input, text, maximum 28 characters>	<<user input, text, maximum 28 characters>	<<User selects from list: Cellulose, Cement EPS compound, Expanded polystyrene, extruded polystyrene, Fiberglass, Insulation other, Mineral wool, Plastic polyisocyanurate, Plastic polystyrene, Polyurethane, Spray Foam Insulation (SPF), Urea formaldehyde>>	<<pull from CF1R but user can override with value equal to or greater than CF1R; else NA >>	<<pull from CF1R but user can override with value equal to or greater than CF1R; else NA >>

D. Raised Floor Insulation

<<if CF1R contains raised floor entry, then display this section; else display header and standard this section does not apply message>>

01	02	03	04	05	06	07	08	09
I.D.	Manufacturer & Brand	Framing Material	Framing Size & Spacing	Insulation Type	Cavity Insulation R-value	Insulation Depth (inches)	Exterior Floor Insulation R-value	Concrete Fill
<<pull from CF1R>>	<<user input, text, maximum 28 characters>>	<<pull. From CF1R>>	<<pull from CF1R>>	<<User selects from list: Cellulose, Cement EPS compound, Expanded polystyrene, extruded polystyrene, Fiberglass, Insulation other, Mineral wool, Plastic polyisocyanurate, Plastic polystyrene, Polyurethane, Spray Foam Insulation (SPF), Urea formaldehyde>>	<<pull from CF1R but user can override with value equal to or greater than CF1R >>	<<User Input: Number; else if D07= NA then display NA>>	<<pull from CF1R but user can override with value equal to or greater than CF1R; else NA >>	<<pull from CF1R; else NA >>

E. Slab Floor/Perimeter Insulation (See Section F. for Insulation Requirements for Heated Slabs)

<<IF CF1R CONTAINS SLAB FLOOR ENTRY, THEN DISPLAY THIS SECTION; ELSE DISPLAY HEADER AND STANDARD THIS SECTION DOES NOT APPLY MESSAGE>>

01	02	03	04	05	06	07	08
I.D	Manufacturer & Brand	Floor type	Insulation Type	Insulation Depth (inches)	Insulation R-Value	Vertical Insulation Length (inches)	Horizontal Insulation Length (feet)
<<if available, pull from CF1R; else allow user input text>>	<<user input, text, maximum 28 characters>>	<<pull. From CF1R; else NA>>	<<User selects from list: Cellulose, Cement EPS compound, Expanded polystyrene, extruded polystyrene, Fiberglass, Insulation other, Mineral wool, Plastic polyisocyanurate, Plastic polystyrene, Polyurethane, Spray Foam Insulation (SPF), Urea formaldehyde >>	<<User Input: Number>>	<<pull from CF1R but user can override with value equal to or greater than CF1R >>	<<User input: Number; else NA>>	<<User input: Number; else NA>>

F. Heated Slab Insulation

01	All heated slabs shall be insulated as required by Section 110.8(g). Footings must meet required insulation levels.
02	Insulation shall be installed from the top of the slab, down 16 inches or to the frost line, whichever is greater. Climate zones 1-15 require R-5, and climate zone 16 requires R-10.
03	Alternatively, vertical insulation from top of slab at inside edge of outside wall down to the top of the horizontal insulation. Horizontal insulation from the outside edge of the vertical insulation extending 4 feet toward the center of the slab in a direction normal to the outside of the building in plain view. Climate zones 1-15 require R-5, and climate zone 16 requires R-10 vertical and R-7 horizontal.
The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.	

G. Minimum Mandatory Measures

01	Insulation - 110.8(a): All installed Insulation is certified and listed with Department of Consumer Affairs, Bureau of Household Goods and Services.
02	Insulation - 110.8(b)2: Urea formaldehyde foam insulation is protected by 4 mil polyethylene vapor retarder.
03	Insulation - 110.8(c): Flame spread and smoke density requirements of the California Building Code (CBC) are met.
04	Raised Floor - 150.0(d): All raised wood-frame floor have a minimum R-19 insulation or equivalent U-factor
05	Slab Floor/Perimeter - 150.0(l): Water absorption rate for the insulation material alone without facings is no greater than 0.3%; water vapor permeance rate is no greater than 2.0 perm/inch and is protected from physical damage and UV light deterioration.
06	Above Grade Exterior Wall - 150.0(c)1 & 150.1(c)6: All 2x4 wood-frame walls have a minimum R-13 insulation or equivalent U-factor not exceeding U-0.095402.
07	Above Grade Exterior Wall - 150.0(c)2 & 150.1(c)6: All 2x6 wood-frame walls have a minimum R-20 insulation or equivalent U-factor not exceeding U-0.069074.
08	Roof Deck - 150.0(a)1: All newly constructed attics <u>that are above conditioned space-systems</u> have a maximum area-weighted average U-factor not exceeding U-0.184.
09	Ceiling/Rafter Roof - 150.0(a)2: All wood-frame ceiling have a minimum R-22 insulation or equivalent U-factor.
10	Vapor Retarder – 150.0(g)1: In Climate Zones 1 through 16, the earth floor of unvented crawl space shall be covered with a Class I or Class II vapor retarder, This requirement shall also apply to controlled ventilation crawl space for buildings complying with the Exception to Section 150.0(d).
11	Vapor Retarder – 150.0(g)2: In Climate Zones 14 and 16, a Class I or Class II vapor retarder shall be installed on the conditioned space side of all insulation in all exterior walls, vented attics and unvented attics with air-permeable insulation.
12	Heated Slabs - 110.8(g): All heated slabs shall be insulated as required. - Insulation shall be installed from the top of the slab, down 16 inches or to the frost line, whichever is greater. Climate zones 1-15 require R-5, and climate zone 16 requires R-10. - Alternatively, vertical insulation from top of slab at inside edge of outside wall down to the top of the horizontal insulation. Horizontal insulation from the outside edge of the vertical insulation extending 4 feet toward the center of the slab in a direction normal to the outside of the building in plain view. Climate zones 1-15 require R-5, and climate zone 16 requires R-10 vertical and R-7 horizontal.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

H. Installed Insulation

01	Installed insulation R-values are the same or greater than listed on the CF1R.
02	No gaps or voids between the insulation and framing.
03	No gaps between the sides or ends of batt insulation.
04	Loose-fill insulation must be installed to the minimum installed weight per square foot (density) of the manufacturer's cut sheet for the proposed R-value.
05	Batt insulation is not compressed (no stuffing of the insulation into the cavity) and is installed to its full thickness.
06	Insulation is cut around obstructions such as electrical boxes.
07	Batt insulation is delaminated around all plumbing and electrical lines in ceilings, walls and floors.
08	Band joists are insulated to the same R-value as the wall.
09	In all narrow cavities the insulation shall be cut to fit or filled with expanding foam.
10	Insulation was installed per manufacturer instructions.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

I. Wall Insulation

01	When allowed by manufacturer, Low expanding foam shall be used to fill gaps and voids around windows and doors. If not, the cavity must be air tight and filled completely with insulation. Batt insulation must be cut to width. No stuffing allowed.
02	Installed wall insulation before installing tubs, showers and fireplaces.
03	Electric Panel on walls separating conditioned and unconditioned space are sealed and insulated behind the panel with rigid insulation or expanding foam.
04	All walls of interior closets vented to the outside for HVAC or water heating equipment have the same R-value and air barrier as the exterior walls and ceiling. Doors are insulated and weather stripped.
05	Ducting not allowed in exterior walls unless insulated to R6 or greater and the insulation and duct are not crushed..
06	Corner channels, wall intersections, and double sided shear walls insulated to the required R-value before enclosing the wall.
07	Insulation that does not fill the cavity placed against exterior air barrier.
08	Band joists are insulated to the same R-value as the walls.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

J. Ceiling/Roof Insulation

01	Insulation extends to the outside edge of the exterior top plates and is flush against any ventilation dams/baffles.
02	Insulation is in direct contact with ceiling, so there are no gaps between the ceiling and the insulation.
03	For chimneys and flues, the insulation is in contact with the sheet metal collar.
04	Can lights are covered with insulation to the same depth as required by the CF1R for ceiling insulation. If not, an area weighted calculation is required to be turned in with this compliance document (CF1R-ENV-02).
05	Walkways and mechanical platforms insulated to the same R-value as required for the ceiling. If not, an area weighted calculation is required to be turned in with this compliance document (CF1R-ENV-02).
06	Insulate a soffit by adding an air barrier and cover with insulation, or insulate the entire soffit including floor and walls.
07	Knee walls and skylight shafts are insulated to the wall R-value and in full contact with the interior air barrier. If framing on these surfaces is laid flat batt insulation is cut to fit around the framing. Batt insulation is not allowed to be draped over the framing.
08	Attic access doors insulated to the same R-value as ceiling. The insulation is permanently attached using adhesive or mechanical fasteners.
09	Attic access must be surrounded with a dam at least the same depth as the insulation to prevent loss of ceiling insulation.
10	Batt insulation cut to fit around cross bracings and truss webs in attic.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

K. Raised Floor Insulation

01	Insulation in full contact with subfloor.
02	Insulation hangers spaced at 18 inches or less, insulation hangers must not compress insulation.
03	If netting or mesh is used, the cavity under the floor is filled and in contact with the subfloor.
04	If the basement is conditioned the walls adjacent to the crawlspace must meet minimum wall R-value requirements. This includes framed stem walls, and vertical concrete retaining walls.
05	If access to the crawl space is from the conditioned area, the raised floor must have an airtight insulated access hatch.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

L. Floor Above Grade Insulation

01	Insulation must be in full contact with subfloor if the air barrier is at the band joist at the garage house wall.
02	Insulation hangers spaced at 18 inches or less, insulation hangers must not compress insulation.
03	If netting or mesh is used, the cavity under the floor is filled and in contact with the subfloor.
04	If air barrier is at the perimeter of the garage, below conditioned subfloor, the insulation is placed on the garage ceiling. Perimeter of subfloor is also insulated.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

M. Cantilevered Floor Insulation

01	Insulation in full contact with cantilevered subfloor. Insulation hangers spaced at 18 inches or less, insulation hangers do not compress insulation.
02	If netting or mesh is used, the cavity under the floor is filled and in contact with the subfloor.
03	Sealed Blocking is installed between joists where wall rim joist would be located in the absence of a cantilever. Insulation is placed on both sides of this block.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.

N. Attached Porch Roof Insulation

01	Exterior insulated wall at the intersection of the porch roof is fully insulated above, below and behind the roof line.
02	Where truss framing is used, airtight blocking is at the top and bottom of each wall/roof section and insulated.

The responsible person's signature on this compliance document affirms that all applicable requirements in this table have been met.